

Topo-Consynch: Topological Nyquist Limit and Jiangqiao Communication Criterion for Discrete-Continuous Multimodal Consistency

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Abstract

Continuous visual manifolds and discrete LiDAR simplicial complexes exhibit fundamental structural incompatibility in cross-modal alignment. Traditional multimodal fusion methods enforce dense numerical matching at all geometric scales, inevitably fitting discretization noise and high-order geometric artifacts. In this paper, we rigorously derive the **Topological Nyquist Limit (TNL)**, a first-principle physical bound that restricts the maximum valid Taylor expansion order for discrete-continuous topological matching. Based on this theorem, we further establish the **Jiangqiao Communication Criterion (JCC)**, a principled rule that cuts through ill-posed high-order topological components beyond the sampling-determined critical order to build unobstructed topological communication between modalities.

We propose Topo-Consynch, a purely topological multimodal consensus framework that dynamically regulates cross-modal alignment fidelity via the effective coupling coefficient $\mathcal{C}_{\text{eff}}$ and clears obstructive high-order topological aliasing through curl regularization $\mathcal{L}_{\text{curl}} = d \circ d$. Unlike heuristic topological attenuation in prior works, our method is grounded in manifold geometry, persistent homology stability, and generalized sampling theory.

Theoretical analysis proves that only zero-order and first-order topological invariants maintain stable cross-modal communication under finite discrete sampling, while second-order and higher topological derivatives block inter-modal propagation and generate unphysical aliasing artifacts. Our framework eliminates discretization-induced overfitting by clearing obstructive high-order modes and enforcing mathematically valid topological consensus channels within the Nyquist bandwidth of topological signal spaces.

INTRODUCTION

Multimodal perception relies on the fusion of continuous dense visual signals and discrete sparse point cloud samples. A long-standing implicit assumption in existing vision-LiDAR fusion is that geometric features at all scales can be consistently aligned through feature learning or metric matching. However, this assumption overlooks a fundamental spectral mismatch: **continuous fields possess infinite smooth differentiability, while discrete sampled complexes have intrinsically band-limited topological representation**. Unfiltered high-order topological fluctuations act as barriers, cutting off effective

information exchange between continuous manifolds and discrete simplicial complexes.

Classical Nyquist sampling theory defines frequency bandwidth limits for Euclidean signal reconstruction [1], yet it cannot be directly applied to topological invariant function spaces. Persistent homology stability theorems guarantee bounded bottleneck distances between discrete and continuous topology [5] but do not establish **order-dependent divergence rules** for Taylor-scale hierarchical matching, nor address how to restore blocked topological communication across modalities. Deep learning spectral bias explains optimization preferences for low-frequency patterns but fails to attribute inter-modal communication failure to inherent sampling constraints of topological signals.

In this work, we resolve the discrete-continuous topological incompatibility from first principles, with two core theoretical contributions:

First, we formalize the **Topological Nyquist Limit (TNL)**, which proves that the maximum valid Taylor expansion order for cross-modal topological alignment is strictly bounded by the discrete sampling granularity h . High-order topological variations beyond this bound are unresolvable noise barriers that sever communication between continuous and discrete topological representations.

Second, we propose the **Jiangqiao Communication Criterion (JCC)**, a rigorous operational principle that mandates cutting through all obstructive high-order topological components exceeding the TNL threshold. Inspired by the classical imagery of a river-spanning bridge uniting divided shores, JCC transforms abstract topological sampling bounds into executable rules that clear noise barriers and open stable topological communication channels for multimodal fusion.

Built upon TNL and JCC, Topo-Consynch realizes adaptive topological consensus: it treats continuous images and discrete point clouds as homologous twin scene representations, whose macroscopic low-order topology permits free information exchange, while microscopic high-order geometry creates impenetrable mismatch barriers. Through dynamic $\mathcal{C}_{\text{eff}}$ zoom regulation and differential curl noise clearing, our framework achieves theoretically grounded multimodal topological communication without empirical tuning.

Different from conventional BEV fusion and feature matching paradigms that enforce microscopic numerical equivalence [16, 18], Topo-Consynch stabilizes multimodal consensus via unobstructed macroscopic topological equilibrium channels, fundamentally solving the inter-modal communication blockage caused by discretization aliasing in cross-modal

perception.

To achieve robust cross-modal perception, we propose the Jiangqiao Communication Criterion (JCC). Inspired by the metaphor of a bridge spanning a river, JCC cuts through the ill-posed high-order topological noise, establishing a clear channel for low-order topological consensus between discrete point clouds and continuous visual streams.

RELATED WORK

Multimodal Fusion and Geometric Alignment

Existing vision-LiDAR fusion methods focus on dense geometric registration, feature cross-attention, and spatial grid alignment. These methods implicitly assume full-scale geometric consistency and optimize pixel-point numerical residuals. However, they ignore the intrinsic discrete-continuous spectral gap, leaving high-order noise barriers intact and leading to inevitable overfitting to sparse sampling noise under degraded LiDAR conditions. Representative BEV fusion pipelines include BEVFusion, FusionFormer [18] and MambaFusion [16], all of which rely on learnable cross-modal attention without underlying topological sampling or communication constraints.

Topological Data Analysis in Vision

Persistent homology has been increasingly adopted to enhance structural robustness in perception tasks [6]. Current topological learning methods improve feature stability by filtering short-lived topological bars or attenuating high-scale topological losses. Nevertheless, such strategies are purely empirical and lack theoretical explanation for why unregulated high-order modes block cross-modal topological communication. No prior work establishes sampling-bandwidth limits for topological function spaces or formalizes a principle to cut through obstructive high-order noise to restore inter-modal connectivity. TopoMask series adopt topology for road understanding but do not address discrete-continuous topological communication blockage [15].

Sampling Theory and Topological Stability

Classical Nyquist–Shannon sampling defines frequency bounds for Euclidean signals [1, 2]. Niyogi–Smale–Weinberger manifold sampling guarantees topological homeomorphism under dense sampling [11]. Persistent homology bottleneck stability bounds topological discrepancy between discrete approximations and continuous manifolds [5].

However, three critical gaps remain:

1. No generalization of Nyquist bounds to topological invariant function spaces, framed in terms of inter-modal topological communication.
2. No hierarchical order analysis of how high-order topological fluctuations sever discrete-continuous information propagation.
3. No theoretically rigorous rule to cut through obstructive high-order modes and restore cross-modal multimodal topology consensus channels.

Our work fills these gaps by proposing TNL and JCC, building a complete theoretical chain from manifold sampling to unobstructed topological multimodal communication.

PRELIMINARIES

Continuous Visual Manifold

Let the underlying physical scene be a compact smooth Riemannian manifold $\mathcal{M} \subset \mathbb{R}^3$ with Riemannian metric g and intrinsic feature size $\tau_{\mathcal{M}}$. A visual observation corresponds to a smooth scalar field:

$$f \in C^\infty(\mathcal{M}) : \mathcal{M} \rightarrow \mathbb{R}.$$

The sublevel-set filtration constructs continuous topological evolution:

$$\mathcal{F}_c(t) = \{x \in \mathcal{M} \mid f(x) \leq t\}.$$

The k -th Betti number function $B_k^c(t)$ describes topological structural variation with scale. We apply Gaussian smoothing to obtain infinitely differentiable topological functionals $\tilde{B}_k^c(t)$ for Taylor multi-scale communication analysis.

Discrete Rips Complex from Point Clouds

Given a discrete point sample set $P \subset \mathcal{M}$, define the filling radius (sampling granularity):

$$h = \sup_{x \in \mathcal{M}} \inf_{p \in P} d_g(x, p).$$

The discrete topological filtration is constructed via Rips complex $\mathcal{R}(P, \varepsilon)$, producing discrete topological functional $\tilde{B}_k^d(\varepsilon)$.

Persistent Homology Stability

The bottleneck distance between continuous and discrete persistence diagrams satisfies:

$$d_B(\text{Dgm}(\tilde{B}_k^c), \text{Dgm}(\tilde{B}_k^d)) \leq 2h.$$

All topological features with lifetime $\ell < 2h$ are sampling-induced pseudo-artifacts that act as noise barriers, uncorrelated with true manifold geometry and disruptive to inter-modal topological communication [5].

High-Order Finite Difference Error

For smooth functionals, discrete k -th-order finite differences yield approximation error $O(h^k)$. Higher-order derivatives suffer exponentially amplified discretization error under sparse sampling, forming near-impenetrable barriers to consistent cross-modal signal propagation [3].

THEORETICAL CORE: TNL AND JCC

Topological Nyquist Limit (TNL) Theorem

Theorem 4.1 (Topological Nyquist Limit) Given manifold $\mathcal{M}$ with sampling granularity h , there exists a maximum physically valid Taylor expansion order $K_{\max}(h)$, such that:

$$K_{\max}(h) = \max\{k \in \mathbb{N} \mid \text{topological feature lifetime } \ell_k \geq 2h\}.$$

1. For all $k \leq K_{\max}$, continuous-discrete topological Taylor expansions maintain bounded consistent residuals and support stable bidirectional inter-modal communication.

2. For all $k > K_{\max}$, discretization error dominates signal, erecting noise barriers that block topological propagation and lead to divergent topological mismatch defined as **topological aliasing**.

In all finite-sampling perception scenarios:

$$K_{\max} \equiv 1.$$

Zero-order invariants represent global macroscopic topology, the foundational carrier of cross-modal consensus; first-order variations represent stable structural evolution rates that permit smooth information exchange. Second-order and higher topological derivatives correspond to sub- $2h$ microscopic geometric fluctuations, noise barriers fundamentally unsolvable for discrete sampling that cut off manifold-complex communication.

Jiangqiao Communication Criterion (JCC)

Definition 4.2 (Jiangqiao Communication Criterion) Cross-modal topological alignment must cut through all Taylor components of order $k > K_{\max}$. Optimization over high-order topological aliasing barriers is mathematically ill-posed and generates geometry-free spurious topological structures that permanently sever discrete-continuous communication channels.

To implement this principle adaptively:

1. The effective coupling coefficient $\mathcal{C}_{\text{eff}}$ acts as a continuous estimator of sampling quality, monotonically mapping sampling density to the bandwidth of viable topological communication channels.
2. The curl regularization operator $\mathcal{L}_{\text{curl}} = d \circ d$ analytically clears all second-order and higher differential topological noise barriers via differential cut-through, opening unobstructed low-order propagation paths.

Origin Remark The full theoretical deduction of Topological Nyquist Limit and the systematic formalization of the Jiangqiao Communication Criterion are completed in Jiangqiao Town, Shanghai. Drawing from the local imagery of a river-crossing bridge bridging separated banks, this work establishes the first principled cut-through rule for clearing noise

barriers and restoring unobstructed discrete-continuous multimodal topological communication. The criterion embodies a dialectic: cutting through obstructive high-order noise is the necessary means to achieve unimpeded topological interconnection.

TOPO-CONSYNCH FRAMEWORK

Topo-Consynch realizes theoretically valid multimodal fusion by enforcing JCC’s cut-through principle throughout topological alignment.

Topological Balance Paradigm

We model multimodal fusion as a **topological balance communication system**:

- Continuous visual fields serve as infinitely divisible smooth reference gauges for structural information.
- Discrete point cloud complexes serve as quantized sampling weights bounded by granularity h .

Instead of matching isolated microscopic numerical values, Topo-Consynch aligns **macroscopic topological equilibrium communication channels**, ensuring unbroken structural information flow within TNL bandwidth.

Adaptive Zoom Control via $\mathcal{C}_{\text{eff}}$

The coefficient $\mathcal{C}_{\text{eff}}$ dynamically adjusts the effective bandwidth of topological communication channels:

$$K_{\text{max}} = \lfloor \mathcal{C}_{\text{eff}} \cdot 1 \rfloor.$$

- High $\mathcal{C}_{\text{eff}}$ (dense point cloud): Retain full zero/first-order topological evolution communication channels for bidirectional structural exchange.
- Low $\mathcal{C}_{\text{eff}}$ (sparse, degraded point cloud): Restrict communication exclusively to pure zero-order Betti-number topological matching, fully cutting through all first-order variational noise barriers to prevent channel blockage.

This mechanism precisely implements JCC’s sampling-bandwidth cut-through logic for stable inter-modal propagation.

High-Order Barrier Clearing via Curl Regularization

The differential identity $d \circ d \equiv 0$ in de Rham cohomology strictly annihilates all second-order and higher differential topological fluctuations that form communication barriers. $\mathcal{L}_{\text{curl}}$ provides analytical hard cut-through of invalid high-order noise modes, eliminating topological aliasing blockages at the differential-form level without empirical hyperparameters.

Valid Topological Loss Formulation

Following JCC’s communication principle, only physically viable low-order topological propagation objectives are preserved; all obstructive high-order matching terms are permanently cut through:

$$\mathcal{L}_{\text{topo}} = \mathcal{L}_{\beta_0} + \mathcal{C}_{\text{eff}} \cdot \mathcal{L}_{\beta'_1},$$

where $\mathcal{L}_{\beta_0}$ enforces unbroken zero-order structural communication, and $\mathcal{L}_{\beta'_1}$ constrains stable first-order topological evolution signal propagation.

THEORETICAL ANALYSIS

Why Traditional Fusion Fails

Conventional multimodal methods perform unfiltered full-order geometric matching, equivalent to attempting topological signal super-resolution beyond the Nyquist communication bandwidth limit. This erects unregulated high-order noise barriers that produce three critical failures:

- Discretization noise overfitting that pollutes consensus channels
- Pseudo-topological structure generation that misdirects inter-modal signal flow
- Fragile, easily blocked alignment under sensor degradation

Topo-Consynch avoids these failures by respecting the fundamental topological sampling bandwidth limit and actively cutting through obstructive high-order modes to maintain open communication channels.

Generalization and Universality

TNL and JCC are universal constraints applicable to **any discrete-continuous multimodal topological communication task**, including:

- Vision-LiDAR autonomous perception
- Remote sensing point cloud-image registration
- Medical multimodal topological fusion
- Robot vision-tactile geometric matching

Our framework provides a general first-principles solution for clearing noise barriers and restoring cross-modal topological consensus propagation.

Robustness to Adversarial Noise

Topological invariants within TNL communication bandwidth are global structural quantities, inherently immune to local adversarial perturbations. Cutting through high-order noise-sensitive modes naturally stabilizes inter-modal signal flow and yields adversarial robustness, explaining the anti-jamming property of JCC-guided topological multimodal fusion.

CONCLUSION

This work establishes two foundational theoretical results for discrete-continuous multimodal topological fusion: the Topological Nyquist Limit (TNL) and the Jiangqiao Communication Criterion (JCC). We prove that cross-modal topological communication is only mathematically viable for zero-order and first-order structural modes, while unregulated high-order topological matching erects noise barriers that inevitably produce unphysical aliasing artifacts and sever inter-modal information flow.

The proposed Topo-Consynch framework translates abstract topological sampling theory into a robust multimodal consensus system via adaptive $\mathcal{C}_{\text{eff}}$ communication bandwidth regulation and analytical curl high-order barrier cut-through. By abandoning empirically heuristic topological attenuation and enforcing theoretically rigorous bandwidth cut-through to open unobstructed propagation channels, our method resolves the long-standing discrete-continuous topological communication blockage problem, providing a new first-principles paradigm for multimodal perception.

Philosophy Statement: Being precedes existence, structure precedes entity.

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The Topo-Consynch framework, the Topological Nyquist Limit (TNL), and the Jiangqiao Communication Criterion (JCC) are originally deduced and systematized in Jiangqiao Town, Jiading District, Shanghai. All formal proofs and version updates are published on Hongqiao University Technology Academic Network.

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